# Base Building Preventative Maintenance Standards

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To be used with BBM Pricing Template version 7.XX only

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GLOBAL CENTRAL RME
STANDARDS & COMPLIANCE

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HVAC SYSTEMS, AIR CONDENSING
HVAC SYSTEMS, KITCHEN EXHAUST DUCT SYSTEM
HVAC SYSTEMS, EXHAUST FAN UNITS
HVAC SYSTEMS, GAS DARK HEATERS
HVAC SYSTEMS, HUMIDIFIERS
HVAC SYSTEMS, AIR ROOFTOP UNITS-OUTDOOR
HVAC SYSTEMS, SUPPLY FAN UNITS
HVAC SYSTEMS, WATER HEATED SYSTEM
LADDERS, FIXED LADDERS
LADDERS, MOBILE LADDERS
LIFTS, GOODS LIFTS
LIFTS, PASSENGER LIFTS
LIFTS, VEHICLE TO GROUND LIFTS
LIGHT, EXTERNAL
LIGHT, INTERNAL
LIGHT, SAFETY LIGHTING AND SIGNAGE
STORAGE, HIGH RACKS
STORAGE, SHELVING
BUILDING WATER SUPPLY
BUILDING WATER SUPPLY, DOMESTIC HOT WATER HEATERS

## What is new in this release?

- PM's have been refreshed to align with latest OCB changes as part of global alignment program.
- Some PMs have been split to improve tracking and forecasting in APM
- The following objects are associated with new PM Packages as a result of OCB Updates, Global Alignment or splitting activity to improve data granularity

| OBJECT | EC |
|---|---|
| BLDG.EXTSURFACES | AMZ.BLDG.EXTSURFACES |
| BLDG.MEZZANINE | AMZ.BLDG.MEZZANINE.GRP |
| BLDG.SRMS | AMZ.BLDG.SRMS |
| BLDG.STRUCTURE | AMZ.BLDG.STRCTR |
| BLDG.WINDOWS | AMZ.BLDG.WINDOW |
| ELEC.HV.SWGR | AMZ.ELEC.HV.SWGR.GRP |
| ELEC.HV.TX | AMZ.ELEC.HV.TX.GRP |
| ELEC.LV.SWBD | AMZ.ELEC.LV.SWBD |
| FIRE.DET.ASD | AMZ.FIRE.ASP.GRP |
| FIRE.DET.BEAM | AMZ.FIRE.BEAM.GRP |
| FIRE.DET.DUCT | AMZ.FIRE.DUCT.GRP |
| FIRE.DET.FLAME | AMZ.FIRE.FLAME.GRP |
| FIRE.DET.HEAT | AMZ.FIRE.HEAT.GRP |
| FIRE.DET.LINEAR | AMZ.FIRE.LNHT.GRP |
| FIRE.DET.MCP | AMZ.FIRE.MCP.GRP |
| FIRE.DET.RADIO | AMZ.FIRE.RADIO |
| FIRE.DET.SMOKE | AMZ.FIRE.SMOKE.GRP |
| FIRE.DET.SMOKEHEAT | AMZ.FIRE.SMOKEHEAT.GRP |
| FIRE.DET.SMOKEHEATCO | AMZ.FIRE.SMOKEHEATCO.GRP |
| FIRE.FLS.FIREDOOR | AMZ.FIRE.FIREDOOR.GRP |

| OBJECT | EC |
|---|---|
| FIRE.FLS.FIRESLD | AMZ.FIRE.SLD |
| FIRE.FLS.HOLDDOOR | AMZ.FIRE.HOLDDOOR.GRP |
| FIRE.SPHZRD.KITCHEN | AMZ.FIRE.KHOOD |
| FIRE.SUP.SHUTTER | AMZ.FIRE.SHUTTER.GRP |
| FIRE.SUP.SPRK | AMZ.FIRE.SPRK |
| FIRE.SUP.VALV | AMZ.FIRE.VALV |
| FIRE.SUP.WETRISERS | AMZ.FIRE.WET.RISERS |
| FIRE.SUP.WTRTNK | AMZ.FIRE.WT |
| HVAC.CNV | AMZ.HVAC.CNV.GRP |
| HVAC.HMD | AMZ.HVAC.HMD.GRP |
| HVAC.WTR | AMZ.HVAC.WTR.GRP |
| LIFTS.V2G | AMZ.LIFTS.V2G |
| REFR.BLST | AMZ.REFR.BLST |
| REFR.DOORS | AMZ.COOL.DOOR |
| REFR.PLANT.01 | AMZ.REFR.PLANT.GRP |
| REFR.PLANT.CNTRL | AMZ.REFR.CNTRL |
| REFR.RIC | AMZ.REFR.RCHILL |
| REFR.ROOM | AMZ.REFR.ROOM |
| STOR.SHELV | AMZ.SHELV |
| WTR.SUPPLY.HTR | AMZ.WS.HTR.GRP |

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**AMAZON BASE BUILDING MAINTENANCE STANDARD**

GLOBAL CENTRAL RME STANDARDS & COMPLIANCE

| | |
|---|---|
| **Task Title** | HVAC SYSTEMS, AIR ROOFTOP UNITS-OUTDOOR - 9 |
| **APM Object** | HVAC.RTU |
| **APM EC** | 1 AMZ.HVAC.RTU.GRP |
| **BBM Handbook Title** | Air Rooftop Units (RTUs) |
| **General tasks always required** | Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc.<br>Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required<br>Add notes for each observation and create a FWO if necessary to eliminate defects<br>Use appropriate tools, chemicals and industry standard practices (RME policy permitting)<br>Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure<br>These tasks are based on industry guidance, SME knowledge and HIE learning. Please ensure vendor specific requirements are followed.<br>To suggest improvements please raise a SIM<br><br>SIM - Link to SIM template |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 **Last Update: (Version)** | Jul-23 | **Geo Applied:** | Intl APM Instance | **F-Gas** | Yes | 3 **Activies may be covered by No-Go restrictions:** | YES | **Special Certificaion Required:** | Hygiene certification required in some countries (e.g. VDI 6022) | **Delivered By:** | Carried out by trained on site technician certified in the electrical field | 4 **OCB:** | BBM |
| 5 **Evidence for CT:** | YES | **Pricing template Position:** | 9 | **Related CT Areas:** | CT.AIR.INDOOR<br>CT.HVAC.12KW<br>CT.HVAC.FGAS | **Assets included in this PM:** | ROOFTOP UNIT OUTDOOR | 6 **Schedule With:** | HVAC.HMD<br>HVAC.BLR<br>(if installed) | | | | |

**Description of System / Technology (According to BBM Handbook)**

7 Roof Top Units (or RTUs) are a packaged unit that is a typical HVAC system type seen across Amazon FC's. RTUs contain all the components needed to provide conditioned air in one concise unit with a single unit being able to provide comfort cooling, heating, supply of an adjustable volume of fresh air, as well as free cooling.

**Additional Information**

8 All sites with the above listed Base Building Asset should conduct routine maintenance and inspections according to this standard PM package. The Amazon maintenance standard supports compliance, reduces safety risks,improves asset reliability & efficiency.
Some maintenances captures several assets within one PM package -> Ensure that corrective/reactive maintenance for assets where proactive maintenance is contained under another asset is created against the original asset.
Note: the yearly PPM should be planned before the summer season starts.
Refrigerant part -> Ensure that leakage and any other requirements according to F Gas Regulation are followed

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 97 | Refrigerant Part Compressor & Safety | Lubricate all moving parts and bearings if reachable | | | | | | PM | | PM | | | |
| 98 | Final Check | Check the O&M manual and carry out additional required tasks | | | | | | PM | | PM | | | |

9 **Change Log**

July 2023 - First issue in new format. Simplification of task scope, alignment with CT.
June 2024 - Moved "Humidifier tasks" to HVAC.HMD PM.

(See imagine on previous page)

1. **APM EC** has been added to each plan to assist planners to more quickly identify where the PM package is allocated
2. **Geo's applied** has been added to indicate which APM Instance or regions a plan is available and/or deployed.
3. **F-Gas** flag has been added to identify systems which may be linked to F-Gas compliance
4. **Source OCB** has been added as some PM's may be based on technologies outside of the BBM OCB
5. **Pricing Template Position** has been added to each PM to assist vendors to identify quickly which discipline the PM relates to on the BBM Vendor Pricing Template
6. **Schedule With** has been added to indicate Object Codes of equipment which might logically have their maintenance scheduled at the same time due to similar locations or disciplines. These are cross referenced to promote use of efficient maintenance scheduling. This is a guide and intended to support planners.
7. **Description of System** has been included and updated to mirror the BBM Handbook
8. **Additional Information** was previously under a different heading and has been moved and renamed.
9. **Change Control** section has been added to each PM page to show the incremental changes made to each PM over time.

## Using this document

This is a living document and whilst updates are minimised, this document (and each PM within it) should be considered living. It is important to avoid taking off-line copies unnecessarily and to actively check the 'Last Update' date field for version control.

To suggest improvements to these standards, please use the appropriate links below:

**For issues relating to content of standards:**
Raise SIM here

**For APM related issues**
River Link and Path below
https://river.amazon.com
APM Help Center --> Select Region --> Base Building Maintenance (BBM) --> Standard PM's

## Definitions & References

**APM** – Asset Performance Management, Amazons Computerised Maintenance Management System (CMMS)

**APM Instance** – APM operates in two 'Instances', one [AMER] covering North America and LATAM, the other [Intl] covering all remaining regions around the world.

**CMMS** – Computerised Maintenance Management System [See APM]

**EC -** equipment configuration acts as a profile for equipment, and populates data (Manufacturer, Model, Revision, Class and Category) linked with it to the equipment created. Standard equipment-specific data is linked from the configuration such as Documents, parts associated and PM Schedules.

**FWO** – Follow-up Work Order (corrective activity)

**HIE** – High Impact Event

**HRO** – High Risk Operation as defined by WHS

**OCB** – Object code builder

**Nesting -** Nesting is simply an indication that higher frequency tasks are included (nested) in lower frequency tasks. In practice, this would mean, that all tasks that are (example) scheduled monthly, will also be undertaken in the annual maintenance avoiding the need for 2 (or more) work orders to be issued concurrently. APM (The CMMS) will automatically accommodate this releasing only the work order with the higher interval.

**No-Go –** No-Go Activity as defined by WHS

**PM – Preventative Maintenance – (See also SC)** Preventative Maintenance may (unlike SC) be intrusive or require specialised training, tooling, certification or perimetry etc
Most appropriate delivery mechanism will be subject to local review and may require SOP, task risk assessment, permits or delivery via sub contracted resource as appropriate. Note:- PM does not automatically indicate that activity cannot be self-delivered. If one or more of the task items within a maintenance activity is determined to be 'PM', the whole maintenance activity will be classified as 'PM'.
Corrective work identified during a PM must be raised as a Follow-up Work Order (FWO)

**SC – System Check (See also PM)** - a sub-type of PM are 'non-intrusive' checks or activities and will generally be self-delivered by technicians with appropriate information, instruction, training and supervision (which may include SOP) but are usually not intended not to require additional specialist knowledge, certification.
Note - This is guidance only and local implementation should be subject to local review to ensure safety at all times.
Corrective work identified during a SC must be raised as a Follow-up Work Order (FWO)

**SME** – Subject Matter Expert

**SOP** – Standard Operating Procedure

**VSOP** – Visual Standard Operating Procedure

**Acknowledgements**

This document brings together practices from industry standards, manufacturer recommendations, compliance standards, learnings from HIE/COE and SME input.

Thank you to all who contribute to continue to continue to improve and refine these maintenance activities which play an important part in standardising our approaches across Amazon.